

1ACC0221 – BIG DATA

Sección: 18516

Program: Computer Science

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AI Repository Discovery

Data sources, dataset construction and exploratory analysis

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2026

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1. Project Proposal

This project is developed within the ecosystem of open-source scientific artificial intelligence tools. This domain was selected due to the accelerated and continuous growth of public repositories related to models, libraries, frameworks, and research-oriented tools. According to GitHub's Octoverse 2024 report, during 2024 developers created more than 70,000 new public and open-source generative AI projects, while total contributions to this type of project increased by nearly 60%. This growth reflects a highly dynamic, broad, and increasingly difficult environment to explore in a structured way.

In this context, the main problem is that the AI software ecosystem is growing in an explosive and disorganized manner. Every week, new GitHub repositories appear related to scientific papers, experimental implementations, or applied tools, but there is no structured system that allows users to discover, compare, and understand which repositories are truly influential beyond superficial metrics such as stars or forks. This creates difficulties in identifying similar repositories, recognizing thematic families, and prioritizing relevant tools within an increasingly large universe.

Based on this, the proposed system seeks to answer four specific questions:

- Which AI repositories exist within a specific domain, and which of them are similar to each other?
- To which thematic family or cluster does a given repository belong?
- What other tools should someone explore if they are already using a specific tool?
- Which repositories are structurally the most influential within the ecosystem, beyond their apparent popularity?

The dataset is suitable for the course because it combines free text in descriptions and topics, numerical variables such as stars and forks, temporal variables such as age and recent activity, and implicit relationships between repositories. This combination supports the subsequent stages of the project: representation, clustering, recommendation, and graph analysis.

2. Data Sources and Access

2.1. Source Inventory

Source: GitHub REST API
URL: https://api.github.com
License: Public API under GitHub Terms of Service.
Format: JSON via REST API (search endpoint), processed with PyGithub.
Current size: 3,034 repositories and 15 columns (10 collected fields and 5 derived features), extracted on 30 April 2026. The first iteration of the project used an 86-repository sample, kept in `data/raw/repos_raw.csv`.
Selection criteria: public repositories with more than 50 stars, found through search queries on AI research topics (machine learning, deep learning, NLP, computer vision, LLMs, among others). GitHub limits each search to 1,000 results, so several queries were combined and duplicates removed by `repo_id`.

Sources considered but not used. At proposal stage, Papers With Code and the Hugging Face Hub were considered to enrich each repository with its paper, benchmarks, models and datasets. They were not integrated: every column in the dataset comes from the GitHub API. They remain possible future work.

2.2. Ethics and Access Note

The data used in this project comes from public sources accessible through the official GitHub API, accessed with a personal token only for authentication and rate-limit control. No scraping of private profiles was performed, and no information restricted by third-party authentication was accessed. The extraction is limited to technical metadata from public repositories, such as descriptions, topics, primary language, stars, forks, and activity dates.

No real names, email addresses, private messages, or sensitive content from individual users were collected. Therefore, the risk related to personal data is minimal. The processed artifacts are used for academic purposes within the course.

3. Dataset Design and Construction

3.1. Schema

Column	Type	Description	Key
repo_id	integer	Unique identifier of the repository on GitHub	PK
nombre	string	Full name in `owner/repository` format	Unique
descripcion	string	Short description provided by the author	
lenguaje	string	Main programming language	
stars	integer	Number of accumulated stars	
forks	integer	Number of repository forks	
topics	list	Thematic tags declared by the author	
fecha_creacion	datetime	Repository creation date	
ultimo_commit	datetime	Date of the last recorded push	
url	string	Link to the repository	

3.2. Derived tables and joins

Table A	Table B	Join Field	Description	Status
repos	topics_expanded	repo_id	Topics normalized into one row per repository–topic pair (multi-label binarization), used for TF-IDF, SVD and the graph.	Implemented
repos	papers	nombre	Links each repository with its scientific paper.	Not implemented (future work)
repos	dependencies	repo_id	Links each repository with its declared libraries.	Not implemented (future work)

4. Data Dictionary

Observed values in the 3,034-repository snapshot.

Column	Type	Description	Range or observed values
repo_id	integer	Unique GitHub ID	265,995 – 1,223,170,290
nombre	string	`owner/repo`	Example: `tensorflow/tensorflow`
descripcion	string	Free text provided by the author	4 to 685 characters (median 79); “No description” if empty (3 repositories)
lenguaje	string	Main language	41 distinct values; Python (1,745), Jupyter Notebook (353), Unknown (287), C++ (189)
stars	integer	Superficial popularity	50 – 194,936 (median 2,174)
forks	integer	Number of times it was forked	0 – 75,296 (median 304)
topics	list	Author-defined tags	Between 1 and 20 per repository (median 8)
fecha_creacion	datetime	Creation date	31 Jul 2009 – 2026
ultimo_commit	datetime	Last recorded push	4 Dec 2017 – 30 Apr 2026
url	string	Direct link	Always `https://github.com/...`
dias_sin_commit	integer	Days from last push to the extraction date	0 – 3,069 (median 328)
antiguedad_dias	integer	Days from creation to the extraction date	2 – 6,116 (median 1,741)
num_topics	integer	Tag richness	1 – 20
ratio_forks_stars	float	forks / stars	0 – 2.29 (median 0.136)
stars_por_dia	float	stars / antiguedad_dias	0.02 – 2,420.5 (median 1.37)

5. Exploratory Analysis and Dataset Scale

The key figures of the current dataset are as follows:

Total repositories: 3,034
Total columns: 15
Missingness after cleaning: 0 across all columns (3 empty descriptions were filled with “No description” and 287 missing languages with “Unknown”)
Memory estimate: about 2.4 MB

Stars range: 50 to 194,936, with a median of 2,174. The distribution is strongly right-skewed: 549 repositories (18%) have 10,000 stars or more, while half have fewer than 2,200.

Unique languages: 41. Python is dominant with 1,745 repositories (57.5%), followed by Jupyter Notebook (353) and repositories with no language reported by GitHub (287; documentation or content not classified).

Repositories active in the last 30 days: 870 out of 3,034 (28.7%). 1,461 (48.2%) have not been pushed to for more than a year, which suggests that many research projects are published but not continuously maintained.

Average age: 1,889 days (approximately 5.2 years), so the dataset mixes established tools with recent projects; 2023 is the most frequent creation year.

The charts in section 7 show the distribution of stars (log scale), the ranking of programming languages, the days since the last commit, the activity categories and the creation year. They are generated by notebook 01 and stored in `reports/eda/`.

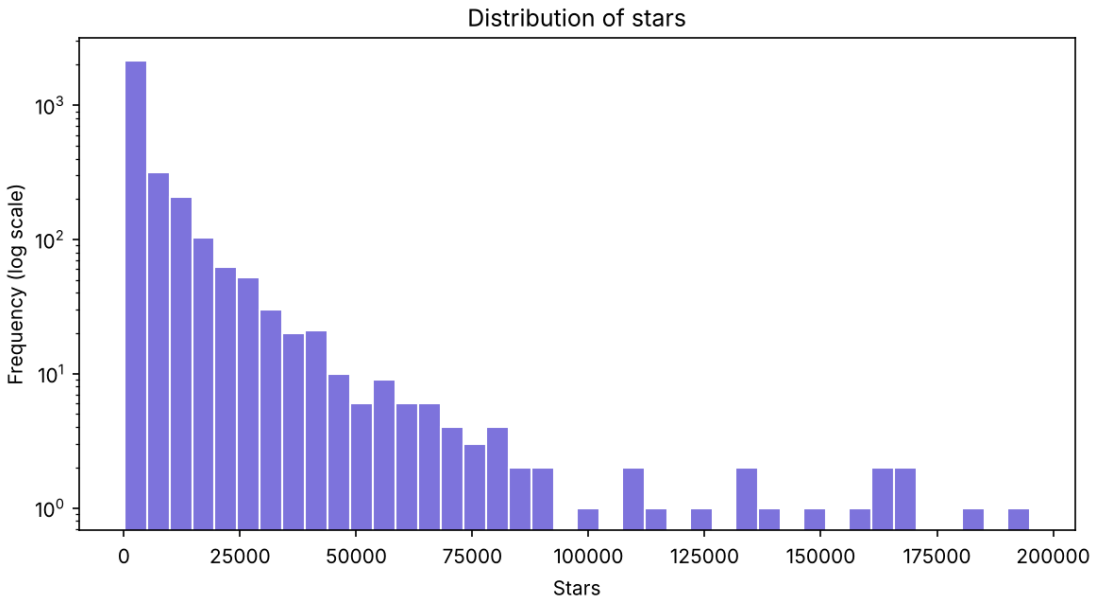
6. Processed Dataset

The processed dataset is generated through a pipeline organized in stages, implemented in `src/ingest.py`:

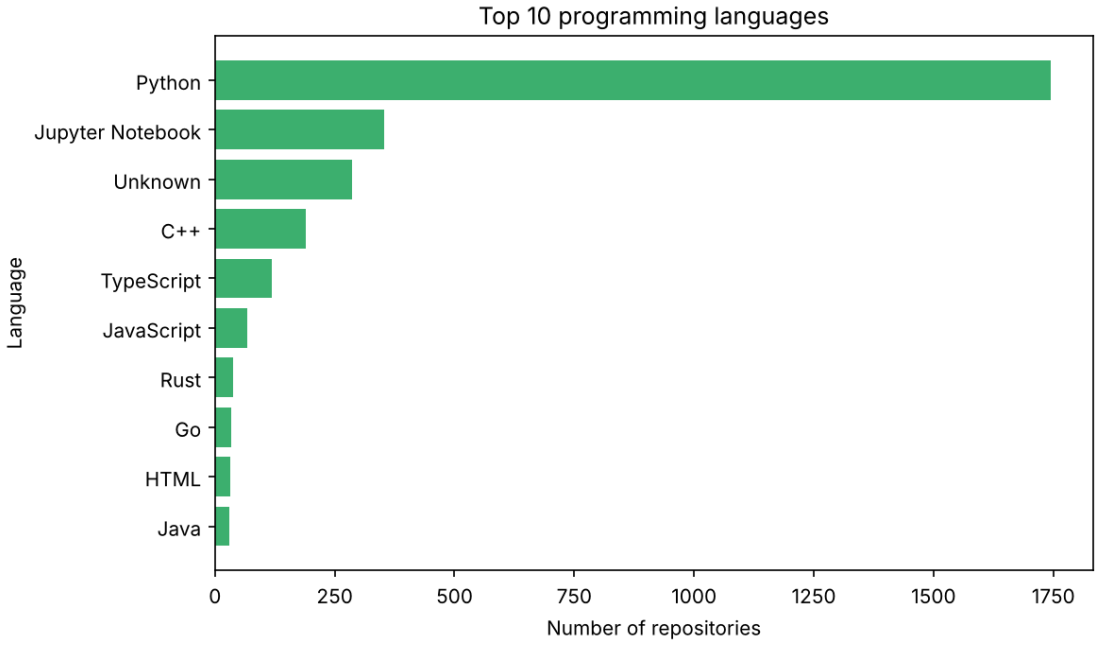
Step 1: the GitHub API is queried to collect the metadata of the repositories defined by the project's search queries.
Step 2: the extraction is stored as a raw table (`repos_raw.csv`). In the repository, `data/raw/repos_raw.csv` keeps the 86-repository sample from the first iteration; new extractions are written to `data/snapshots/<date>/`.
Step 3: the raw table is cleaned: empty descriptions and languages are imputed, dates are converted to UTC, and five features are derived: `dias_sin_commit`, `antiguedad_dias`, `num_topics`, `ratio_forks_stars` and `stars_por_dia`.
Step 4: the result is saved as the processed dataset `data/processed/repos_clean.csv` (the official 3,034-repository snapshot of 30 April 2026, used by every later stage and by the dashboard).

A new extraction can be launched from the project root with `python src/ingest.py`. By default it does not overwrite the official snapshot; `--promote` replaces it. Because stars and activity change daily, a new extraction will not reproduce exactly the numbers reported here.

7. EDA Analysis

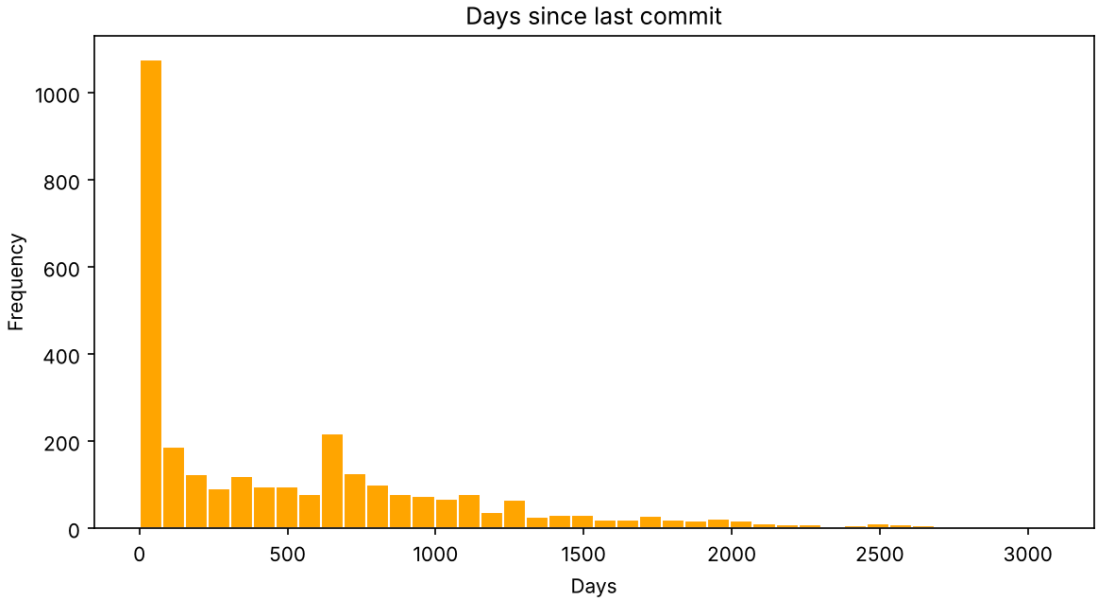


The distribution of stars shows a strong right-skewed pattern (log scale on the vertical axis). Most repositories have low or intermediate popularity, while a small group concentrates very high numbers of stars, up to almost 195,000. Visibility within the dataset is not distributed uniformly.

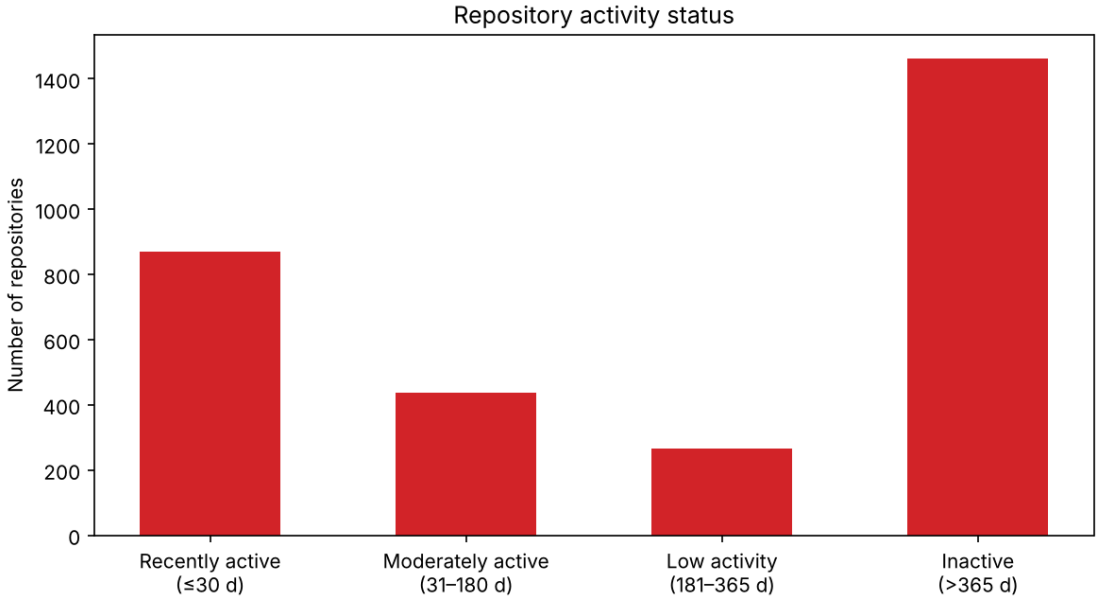


Python clearly predominates, followed by Jupyter Notebook and repositories with no declared language. This is consistent with the ecosystem analyzed, since Python is the most widely used

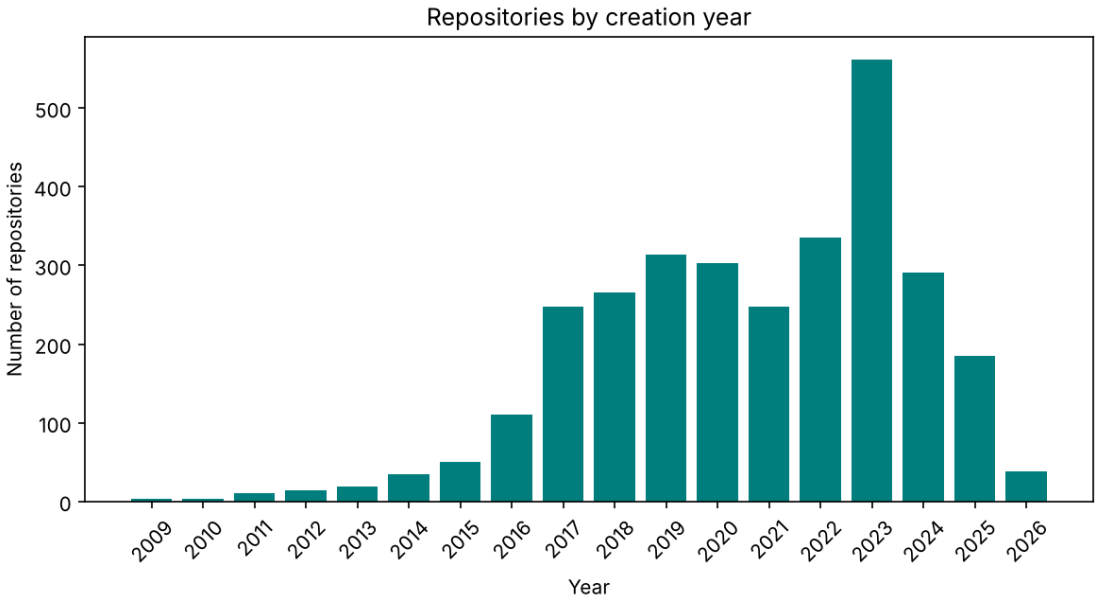
language in artificial intelligence and scientific software. C++ (189), TypeScript (118) and JavaScript (67) follow at a distance.



The distribution of days since the last commit shows a large group of recently updated repositories and a long tail of projects without updates. The dataset includes both actively maintained tools and repositories with limited or discontinued maintenance.



Activity categories use days since the last push: recently active (30 days or less, 870 repositories), moderately active (31–180 days, 436), low activity (181–365 days, 267) and inactive (more than 365 days, 1,461). The predominant category is inactive, so the collected ecosystem combines recognized projects with very different levels of recent maintenance.



Most repositories were created between 2017 and 2025, with 2023 as the peak year (562 repositories). The counts for 2026 are partial because the extraction took place on 30 April.

8. Bibliography

GitHub Staff. *Octoverse 2024: AI leads Python to top language as the number of global developers surges*. GitHub Blog, 2024.